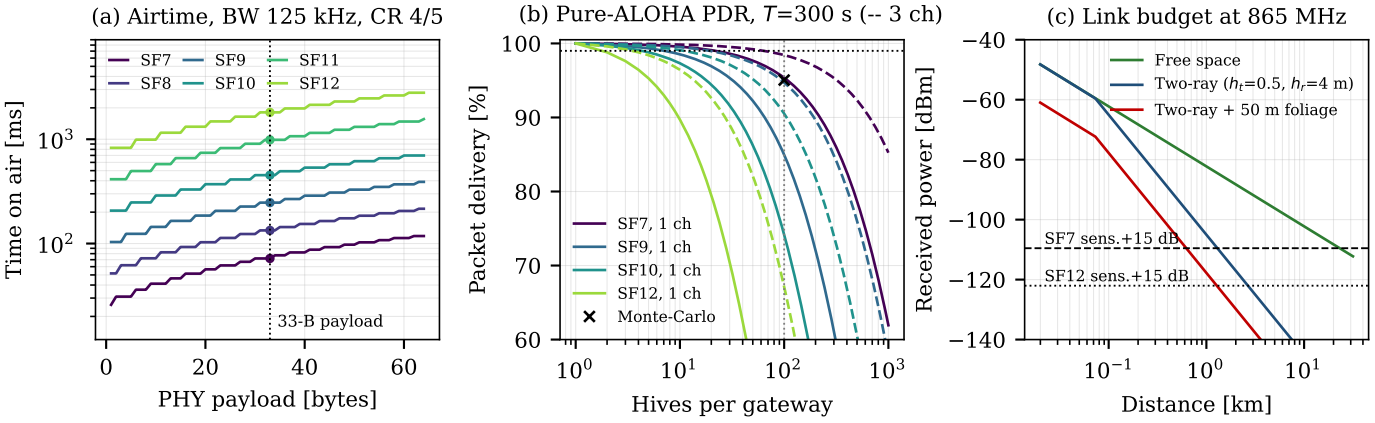
(a) Airtime, BW 125 kHz, CR 4/5
SF7
SF8
SF9
SF10
SF11
SF12
Time on air [ms]
10^3
10^2
33-B payload
0
20
40
60
PHY payload [bytes]
(b) Pure-ALOHA PDR, T=300 s (-- 3 ch)
Packet delivery [%]
100
90
80
70
60
SF7, 1 ch
SF9, 1 ch
SF10, 1 ch
SF12, 1 ch
Monte-Carlo
10^0
10^1
10^2
10^3
Hives per gateway
(c) Link budget at 865 MHz
Received power [dBm]
−40
−60
−80
−100
−120
−140
Free space
Two-ray (h_t=0.5, h_r=4 m)
Two-ray + 50 m foliage
SF7 sens.+15 dB
SF12 sens.+15 dB
10^{-1}
10^0
10^1
Distance [km]